

Confirmed Friction

5:00 – 5:30

Transcript excerpt:
[T] No matter what the size of the angles are, if you have a five pointed star inscribed in a circle it will be 180 degrees.

Heuristic: RISKY (low quality)

ELECTRA: RISKY (No Move → low)

Human label: ✓ FRICTION

LLM fusion:
friction=True
"Teacher states the answer students should discover."
Alt: Ask students to share their measured result first.

Heuristic False Positive

21:30 – 22:00

Transcript excerpt:
[T] Just make sure – just make sure they join. Get rid of that one. Use the pointer tool, just to make the shape.

Heuristic: RISKY (high pressure)

ELECTRA: RISKY (Pressing for Accuracy)

Human label: ✗ Not friction

LLM fusion:
friction=False
"Pressure refers to software construction steps, not mathematical reasoning."

Annotator Disagreement

8:00 – 8:30

Transcript excerpt:
[T] Or they could be words that maybe you do know, but you think that's a pretty important word. Just do this with a brief discussion.

Heuristic: RISKY (unknown — no keyword)

ELECTRA: HIGH (Pressing for Reasoning)

Human label: ✗ Not friction

LLM fusion:
friction=False
"Teacher prompts collaborative vocabulary discussion — a high-quality instructional move."